JOSÉ I. CAIZA

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EDUCATION

Purdue University – Ph.D. in Electrical and Computer Engineering 2023 – July 2027 (expected)
- Advisors: Dr. Philip E. Paré and Dr. Junjie Qin
- Thesis Title: *Distributed Optimization and Learning of Layered Networked Dynamical Systems*

Purdue University – M.Sc. in Electrical and Computer Engineering 2021 – 2023
- Thesis: *Optimal Ordering Policies for Supply Networks with Disruptions*

Escuela Superior Politécnica del Litoral – B.Sc. in Electrical Engineering 2019
- Senior Thesis: *Implementation of a Cascaded Automatic Control with PLCs*

AWARDS

- **Perry Graduate Teaching Fellowship**, Mechanical Engineering, Purdue, Fall 2026
- **Graduate Student Fellowship**, Lawrence Berkeley National Laboratory, June – August 2024
- **Fulbright Foreign Student Award**, USA Department of State, 2021 – 2023
- **Graduate Tuition Scholarship**, Purdue University Graduate School, 2021 – 2023
- **Best Industrial Electronics & Control Systems Award**, IEEE Ecuador Technical Chapters Meeting (ETCM), 2019

PUBLICATIONS

Journals:

- **J. I. Caiza**, J. Qin, P. E. Paré, "Disruptions in Multi-stage Stochastic Inventory Control for Decentralized Serial Networks," *IEEE Transactions on Control of Network Systems (TCNS)*, 2026
- S. Gracy, **J. I. Caiza**, P. E. Paré, C. A. Uribe, "Multi-Competitive Time-Varying Networked SIS Model with an Infrastructure Network," *IFAC Journal of Systems and Control (IFAC-JSC): Special Issue on Systems and Control in Biological and Medical Systems*, 2024

Conferences (peer-reviewed):

- S. Samadi, **J. I. Caiza**, S. Gracy, P. E. Paré, "A Networked SIS Epidemic-Opinion Model with Higher-Order Interactions," in Proceedings of the 65th *IEEE Conference on Decision and Control (CDC)*, 2026
- **J. I. Caiza**, J. Qin, P. E. Paré, "Global and Distributed Reproduction Numbers of a Multilayer SIR Model with an Infrastructure Network," in Proceedings of the 63rd *IEEE Conference on Decision and Control (CDC)*, 2024
- **J. I. Caiza**, I. Walter, J. Panchal, J. Qin, P. E. Paré, "Optimal Ordering Policies for Multi-Echelon Supply Networks," in Proceedings of the 61st *IEEE Conference on Decision and Control (CDC)*, 2022
- **J. I. Caiza**, D. Morán, D. Plaza, "Improvements on Disturbances Rejection in Steam Distillation through IMC-based PID Design," in Proceedings of the 4th *IEEE Ecuador Technical Chapters Meeting (ETCM)*, 2019

Under Review:

- **J. I. Caiza**, J. Qin, P. E. Paré, "Local Reproduction Numbers for Analysis and Optimal Mitigation of Epidemic Processes," under review for *IEEE Transactions on Control of Network Systems (TCNS)*, 2026
- **J. I. Caiza**, J. Qin, P. E. Paré, "Network Cluster-based Upper Bounds of the Basic Reproduction Number for Epidemic Mitigation," under review for Proceedings of the *American Control Conference (ACC)*, 2027
- **J. I. Caiza**, J. Wayment, J. Panchal, P. E. Paré, "Product Update Strategies for Clustered Populations: A Basic Reproduction Number Approach," under review for Proceedings of the *American Control Conference (ACC)*, 2027

Working Papers:

- **J. I. Caiza**, J. Qin, P. E. Paré, "Online Network-Constrained Scheduling of Flexible Loads via Lyapunov Optimization: An Application to Data Centers," 2026

RESEARCH EXPERIENCE

Purdue University Aug 2021 – Present
Research Assistant *West Lafayette, IN*

- Characterized a risk-aware optimal threshold policy for a multi-stage stochastic inventory control, quantifying the asymptotic effect of supply chain disruptions on firms' long-term net income
- Designed a dual-decomposition distributed algorithm for resource allocation to control networked epidemic processes, achieving a 30% reduction in computation time under budget constraints
- Implemented a data-driven incremental algorithm to quantify the risk of solar energy integration in large-scale power systems using the Conditional Value at Risk (CVaR) and polyhedral approximations

Lawrence Berkeley National Laboratory June – August 2024
Visiting Researcher *Berkeley, CA*

- Developed Python parsers to process and analyze datasets from two utility companies (∼185k customers each), enabling quantitative modeling of large-scale power distribution networks
- Validated data-driven models of large-scale power distribution networks by modeling system losses in power flow analysis, yielding a 91% model fit between measured data and simulations
- Quantified cost implications of grid expansion scenarios using load forecasts and solar integration through the LODGE model, informing optimal grid planning decisions

Escuela Superior Politécnica del Litoral Aug – Dec 2019
Research Assistant *Guayaquil, Ecuador*

- Applied Ensemble Kalman Filter (EnKF) for dynamic state estimation in non-linear transmission power systems with real-time inputs, achieving a 94% accuracy under measurement uncertainty
- Developed a multi-loop control system for steam distillation process model by using the internal model control to reject disturbances and maintain system stability

TEACHING

Purdue University
Instructor *West Lafayette, IN*

ME 36500 Measurement and Control Systems I Fall 2026

- Taught concepts of dynamical systems modeling, linearization of nonlinear systems, stability of LTI systems, frequency response, and methods for LTI system identification
- Contributed to the creation of weekly homework and exams during the weekly meeting with the senior lecturers/professors of the course

ME 37500 Measurement and Control Systems II Summer 2026, Summer 2025

- Taught concepts of stability of dynamical systems and controller design in the time and frequency domain: direct pole placement, lead-lag controllers (using root locus and Bode plots), sensitivity analysis
- Supervised weekly laboratory sessions and mentored student teams on hands-on final projects involving the design and implementation of a controller for a line-follower robot

Purdue University
Teaching Assistant *West Lafayette, IN*

ECE 60200 Lumped System Theory Fall 2025, Summer 2022

- Taught students during office hours the analysis of state space models covering topics from stability of dynamical systems to observer-based feedback controller design and model order reduction
- Responsible for the creation and distribution of rubrics for homework/exams and their grading

ECE 51032 Computational Methods for Power Systems Analysis Spring 2025

- Held office hours and led students in implementing optimization algorithms for power system analysis
- Graded students' homework and advised student projects based on current challenges in power systems optimization & control

Escuela Superior Politécnica del Litoral 2016 – 2018
Teaching Assistant *Guayaquil, Ecuador*

- Supported instructors by leading office hours to assist students across Digital Control, Control Systems Theory, Electromagnetic Theory, and Linear Algebra
- Prepared lecture exercises and developed monthly quizzes to support students' learning

SERVICE

Conference Organizer

- *Institute for Optimization, Control, and Networks (ICON) Student Research Conference*, Purdue University, 2027

Invited Session Organizer

- **J. I. Caiza**, S. Gracy, P. E. Paré, "Analysis and Control of Complex Networks," *IEEE Conference on Decision and Control (CDC)*, 2024

Peer Reviewer

- IEEE Conference on Decision and Control (2026, 2025, 2024)
- American Control Conference (2026, 2025)
- IEEE Control Systems Letters (2026, 2025)
- IEEE Transactions on Automatic Control (2026)
- IFAC Workshop on Cyber-Physical & Human-Systems (2026)

PRESENTATIONS

Invited Talks

- **J. I. Caiza**, "Optimal Distributed Mitigation of Layered Epidemic Processes," *Mathematical and Statistical Modeling of Spread and Mitigation of Epidemics*, National Science Foundation: Institute for Mathematical and Statistical Innovation, 2026
- **J. I. Caiza**, "Global and Distributed Reproduction Numbers of a Multilayer SIR Model with an Infrastructure Network," *63rd IEEE Conference on Decision and Control (CDC)*, Milan, Italy, 2024
- **J. I. Caiza**, "Optimal Ordering Policies for Multi-Echelon Supply Networks," *61st IEEE Conference on Decision and Control (CDC)*, Cancún, Mexico, 2022

Conference Talks

- **J. I. Caiza**, "Improvements on Disturbances Rejection in Steam Distillation through IMC-based PID Design," *4th IEEE Ecuador Technical Chapters Meeting (ETCM)*, Guayaquil, Ecuador, 2019

POSTERS

- **J. I. Caiza**, "Local Reproduction Numbers in Networked Epidemic Processes: Analysis and Control," *Midwest Workshop on Control and Game Theory (MWCGT)*, University of Illinois Urbana-Champaign, 2025
- **J. I. Caiza**, "Global and Distributed Reproduction Numbers (DRNs) of a Multilayer SIR Model," *Annual Security Symposium, The Center for Education and Research in Information Assurance and Security (CERIAS)*, Purdue University, 2024

INDUSTRY EXPERIENCE

PepsiCo 2020 – 2021

Country Transformation Leader *Guayaquil, Ecuador*

- Designed and planned 31+ projects reducing production costs by $1.7M USD via net present value & return on investment analysis and quantitative decomposition of volume and price effects
- Executed $200K USD in production cost reductions by leveraging data-driven solutions and working jointly with plant managers, Supply Chain COE, and McKinsey & Company consultants
- Developed operations standards using statistical process control and data-driven approaches resulting in a $15K USD net production cost reduction in Q4 2020

SKILLS & ACTIVITIES

Programming Python, MATLAB & Simulink, OpenDSS, Git, Gurobi
Languages Spanish (native), English (proficient)